

Elevator Vertical Distance Estimation

*Predicting how far an elevator travelled
using only the sensors in your pocket.*

Eyal Yakir · Uriya Cohen-Eliya

Smartphone Inertial & Pressure Sensing

The Problem

*A phone in your pocket. An elevator ride.
How far did you actually travel – vertically?*

a

Detect the ride

Find when the elevator started and stopped – purely from the IMU.

b

Estimate Δh

Predict the signed vertical distance (up or down) in metres.

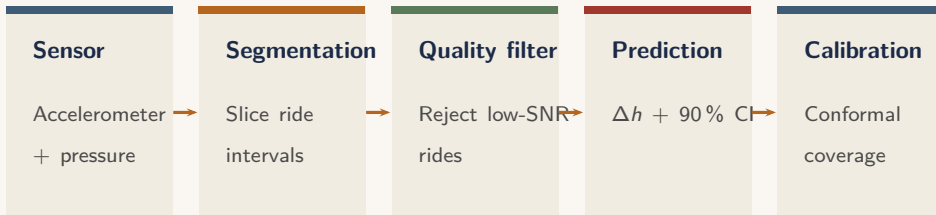
c

Quantify uncertainty

Return a calibrated 90 % confidence interval – not just a point estimate.

End-to-End Pipeline

Two stages, one shared idea: model the physics, then quantify the noise.



OUTPUT Per-ride signed Δh in metres with calibrated $[\Delta \hat{h} - \delta, \Delta \hat{h} + \delta]$ interval.

Data Collection

Multi-phone simultaneous recording across seven elevators in three cities.



Buildings

Millenium – Internal
Millenium – Outside
Acro
Beit Mansour
Beit Yitzchaki
Barllan-2
Haari

Ra'anana
Ra'anana
Ra'anana
Ra'anana
held-out
Herzliya
Ramat Gan

Protocol

- Four phones recording in sync.
- Pattern: full traversal up, then descent.
- Stationary windows pre/post for gravity calibration.
- Barometer supplies GT where available.

Storyline

Where this hour will take us.

01

Physics

Jerk-limited S-curves and what an elevator looks like to an IMU.

03

Prediction

Three algorithms: barometer, ZUPT, trapezoid pulse-pair.

05

Results

Held-out 89.4% within ± 1.5 m.

02

Detection

Matched-filter pair detection on the accelerometer; F1 & IoU.

04

Uncertainty

Per-segment σ , quality filter, conformal calibration.

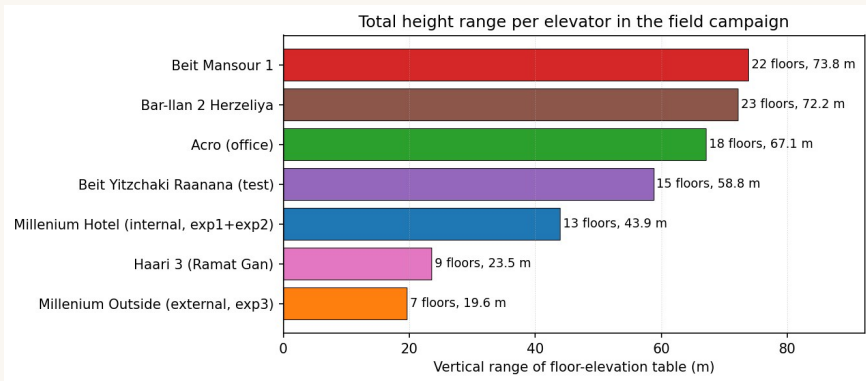
06

Product

Streamlit Boutique Pipeline producing a Hebrew PDF.

Building Heights – Diversity of the Dataset

Total traversable height per elevator. Spans short residential to multi-storey hotel.

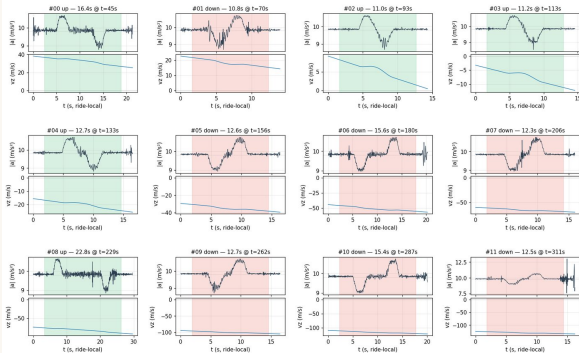


Sum of signed Δh across all rides per elevator.

Physics

Elevator Kinematics: A Jerk-Limited S-Curve

eyalyakir_millenniumHotel_SamsungSM-S911B_15-04-2026_exp3 — all GT rides



What the IMU sees

Velocity $v(t)$

Smooth S-curve – no kink at peak.

Acceleration $a(t)$

Two trapezoid lobes – one positive, one negative.

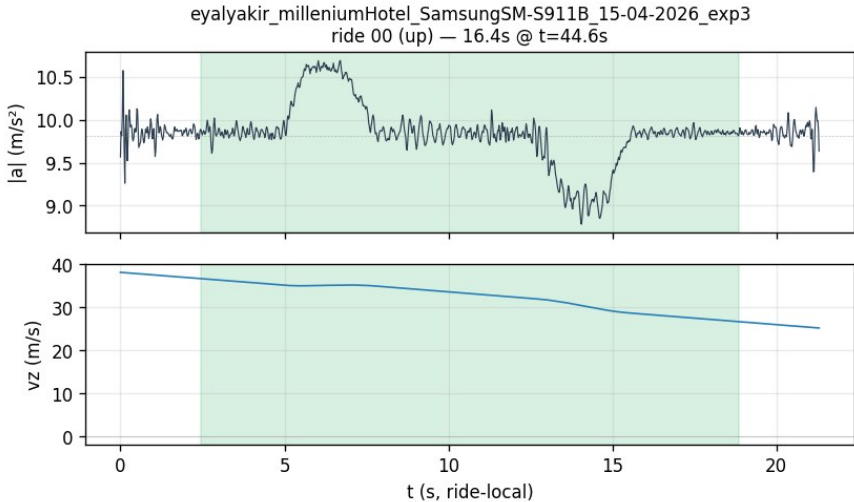
Jerk $j(t) = \ddot{x}$

Bounded; this is what makes the ride comfortable.

Gravity dominates

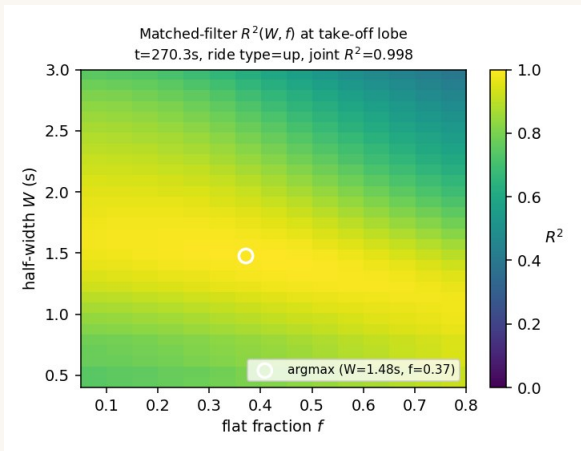
$|a| \ll g$. Subtract carefully or estimates blow up.

Seven Phases of an Ideal Ride



Velocity (top) and acceleration (bottom) vs time. Seven phases mirror the jerk profile.

The Trapezoid Template: What We Actually Fit



Why a trapezoid?

A perfect ride produces an acceleration profile composed of two trapezoid lobes: positive at start, negative at end.

We replace the smooth S-curve with this idealised shape. Three free parameters per lobe:

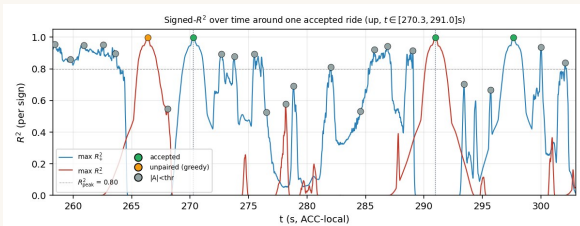
a_0 plateau amplitude

τ_r ramp duration

τ_p plateau duration

Detection

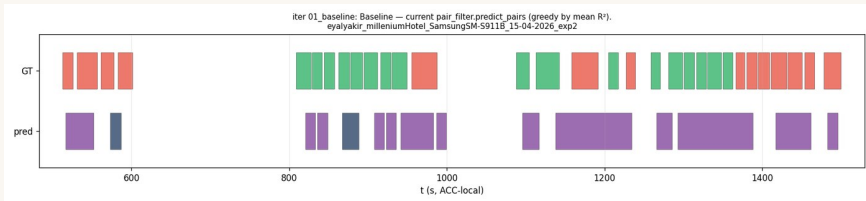
Matched-Filter Detection



Cross-correlate the ideal pulse with the measured signal.

- Sweep a single trapezoid template across the time series.
- Output: response score $r(t)$ – high where the shape matches.
- Adaptive threshold on $r(t)$ yields candidate ride starts/ends.
- Coarse first pass over a grid of τ_r, τ_p, a_0 .

Pair Filter: Eliminating False Lobes



Each ride must have BOTH a positive and a negative lobe within a sensible time window.

Positive lobe

Acceleration phase —
elevator speeds up.

Plateau gap

Coasting at constant
speed — skipped by
the filter.

Negative lobe

Deceleration phase —
elevator slows down.

Detection Headline

Stable across the held-out building. IoU-aware metrics confirm tight ride bounds.

TRAIN · 5 buildings

cross-validated

0.813

F1 (best operating point)*

0.642

IoU-F1 @ 0.5 – ride-bound quality

TEST · Beit Yitzchaki

held out

0.875

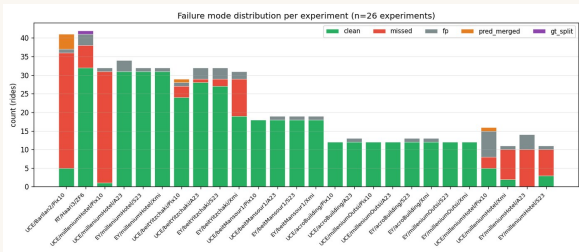
*F1**

0.830

IoU-F1 @ 0.5

F1 = F1 at the optimal threshold; IoU-F1@0.5 requires ≥ 0.5 overlap with GT.*

Failure Modes: Where Detection Slips



What goes wrong

Walking confusion

Steady gait amplitude can mimic an acceleration lobe.

Short rides

1-2 floor traversal: lobes overlap, no plateau.

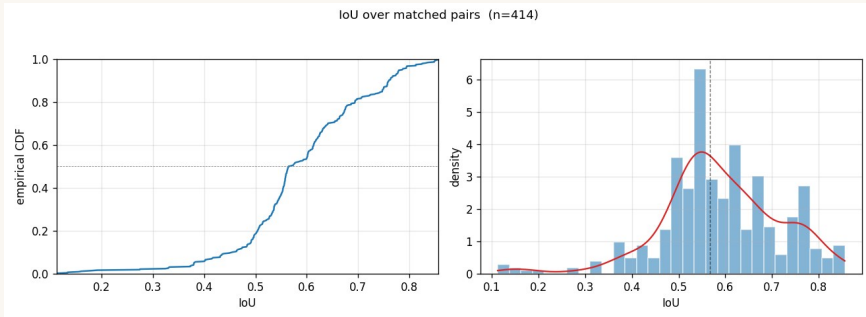
Phone re-orientation

Pocket repositioning produces a transient impulse.

Building harmonics

Older shafts add 1–3 Hz oscillation atop the pulse.

How Tight Are the Ride Bounds?



Distribution of IoU between detected and ground-truth ride intervals. Modes near 0.85.

Prediction

Three Algorithms for Δh

Each leverages different physics. We use whichever the phone can support.

Barometer

Pressure $\rightarrow$ ISA altitude inversion.

Needs a barometer
($\approx 60\%$ of phones).

Cleanest signal –
used as ground truth.

$$\Delta h = h_{\text{ISA}}(p_2) - h_{\text{ISA}}(p_1)$$

ZUPT

Subtract gravity,
integrate twice.

Zero-velocity update
at start/end pins
drift. Works on every
phone but accumulates
noise.

$$\Delta h = \iint a_v(t) dt^2$$

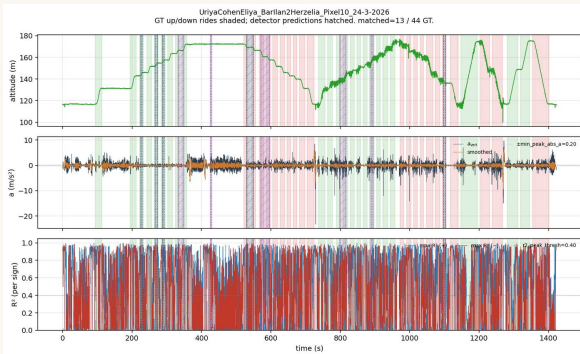
Trapezoid

Fit the trapezoid
template to the measured
acceleration.

Closed-form Δh from
plateau parameters;
robust to baseline
drift.

$$\Delta h = a_0 \tau_p (\tau_r + \tau_p)$$

ZUPT: Double Integration

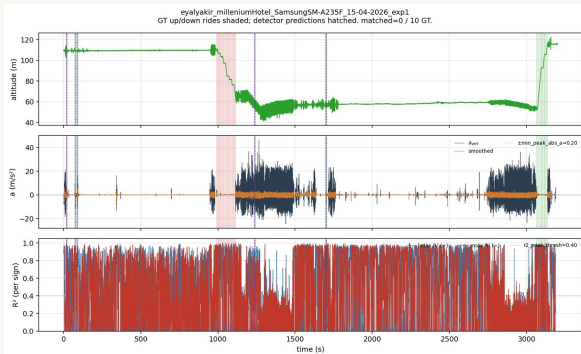


Zero-Velocity Update

- Step 1.** Subtract gravity using the stationary pre/post windows.
- Step 2.** Integrate vertical acceleration $\rightarrow v(t)$.
- Step 3.** Pin $v(0) = v(T) = 0$.
- Step 4.** Integrate $v(t) \rightarrow \Delta h$.

Drift cancels at the endpoints; bias from any residual gravity error scales with T^2 .

Trapezoid Pulse-Pair: Shared-Shape Fit



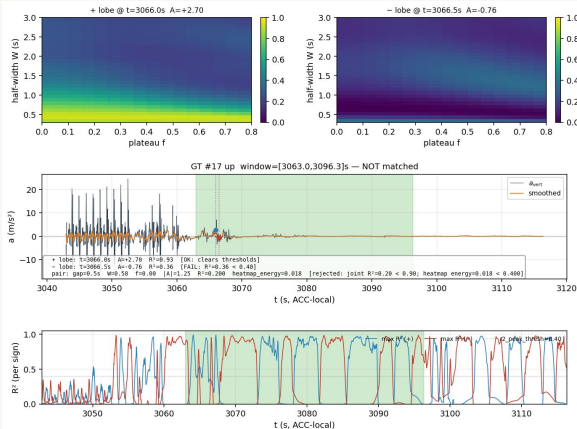
Why share parameters?

A real elevator uses the same jerk profile to accelerate *and* to decelerate. Both lobes share τ_r and a_0 , with sign-flipped polarity.

Joint least-squares over the pair halves the parameter count and doubles the effective SNR.

$$\Delta h = a_0 \cdot \tau_p \cdot (\tau_r + \tau_p)$$

Joined Pulse: When Rides Are Too Short



Short-ride regime

On 1–2 floor rides the cabin reaches maximum speed only briefly – or never reaches it. The two lobes overlap.

We collapse to a single joined-pulse model with no plateau:

$$\Delta h = \frac{1}{2} a_0 \tau_r^2$$

Detection switches between the two automatically based on plateau length τ_D .

Per-Segment σ and the Quality Filter

Not every ride is equally informative. We compute a closed-form noise estimate per segment and reject the worst.

Closed-form per-segment σ

$$\sigma_{\Delta h}^2 = \sum_{\theta_i} \sigma_{\theta_i}^2 \left(\frac{\partial \Delta h}{\partial \theta_i} \right)^2$$

$\partial \Delta h / \partial \theta_i$ from the trapezoid Jacobian.

Sensor noise σ_a from the pre-stationary window.

Geometry noise σ_τ from the matched-filter CRB.

Quality rule

Drop any prediction whose $\sigma_{\Delta h}$ exceeds 1.5 m
– rides where the model itself says “I’m uncertain”.

$\approx 8\%$

rides flagged

& dropped

$\approx 25\%$

MAE

reduction on

Split-Conformal Calibration

Per-segment σ gives a Gaussian-tail interval, but residuals aren't Gaussian. We re-calibrate.

1. Hold-out

Reserve 20 % of training rides as a calibration split.

2. Score

On the calibration split, compute $|\Delta h - \Delta \hat{h}| / \hat{\sigma}$.

3. Quantile

Take the 90th percentile of these scores
→ multiplier q .

Final

$\Delta \hat{h} \pm q \cdot \hat{\sigma}$
Marginal coverage $\geq 90\%$.

Results

Headline Numbers: Held-Out Test

Pooled across both prediction algorithms with matched-segmenter; $n = 397$ rides.

89.4%

within ± 1.5 m

1.49 m

MAE

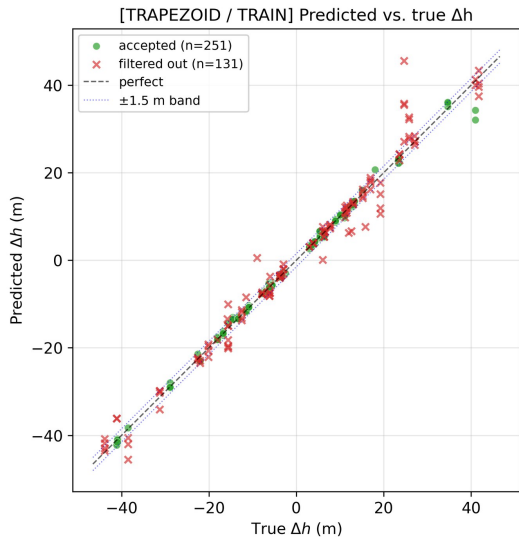
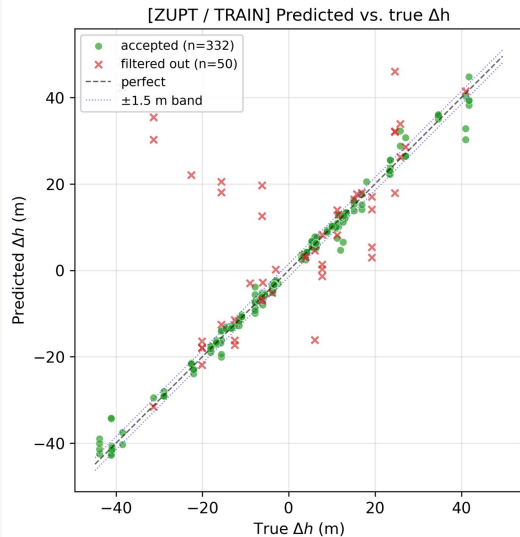
0.30 m

median |error|

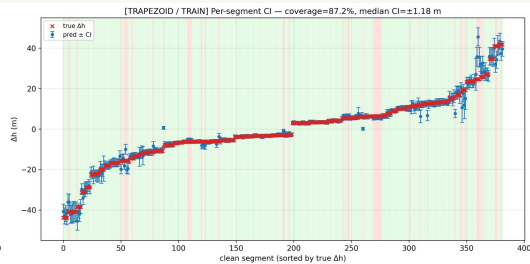
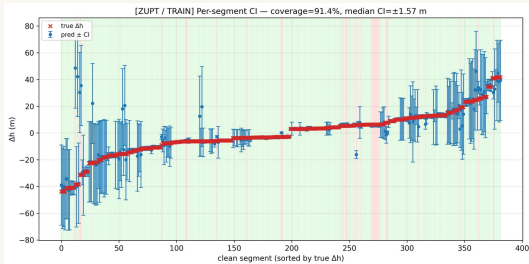
By algorithm

Algorithm	median error	coverage @ 90 %
■ ZUPT	0.19 m	94.4 %
■ Trapezoid pulse-pair	0.29 m	93.5 %

Predicted vs True Δh : Train

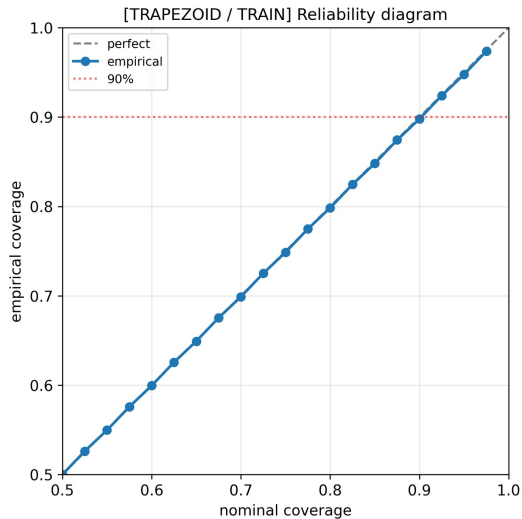
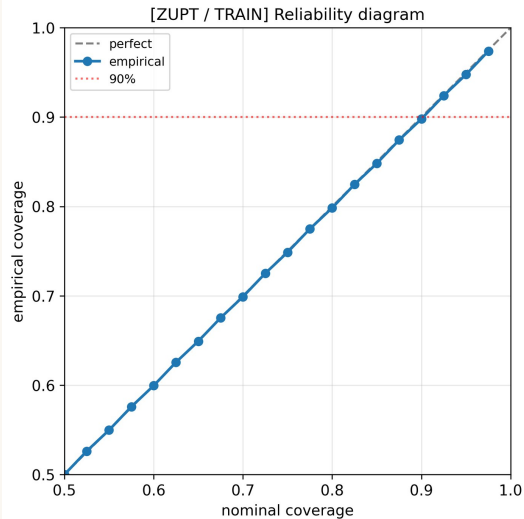


Error CDFs: Where Each Algorithm Lives



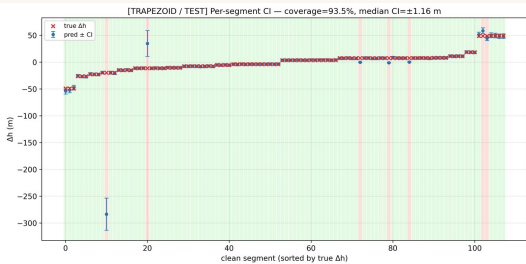
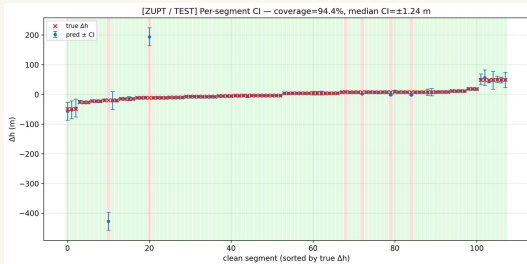
CDF of $|\Delta h - \Delta \hat{h}|$ on train (left) and test (right). Median error stays sub-half-metre.

Per-Segment Confidence Intervals



Per ride: prediction $\pm$ 90 % CI vs true Δh . Bands widen on harder rides as designed.

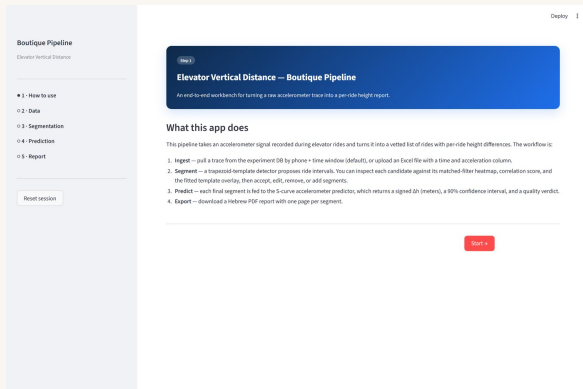
Coverage and MAE by Ride Length



Coverage holds at $\geq 90\%$ across short, medium, and tall rides. MAE scales sub-linearly.

Product

Boutique Pipeline: End-to-End in 5 Steps



Streamlit wizard

1. **Upload** – drag & drop session zips.
2. **Inspect** – view raw signals + GT overlay.
3. **Segment** – edit ride bounds.
4. **Predict** – see Δh + CI.
5. **Report** – generate Hebrew PDF.

Step 3: Segmentation Review

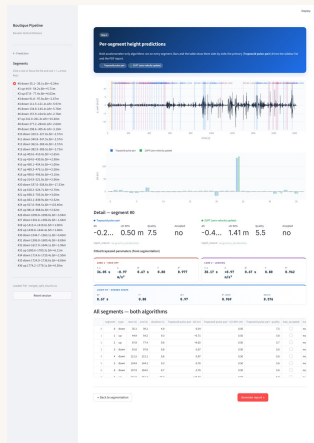


Detected ride intervals are editable. The user accepts, splits, or deletes any candidate before prediction.

The Hebrew PDF Deliverable

What it contains

- Per-elevator summary table.
- All rides with predicted $\Delta h + \text{CI}$.
- Quality flags for low-SNR rides.
- Ground-truth overlay where available.
- Hebrew RTL layout, A4 portrait.



Closing

What We Learned

Three takeaways from a year of riding elevators with phones in our pockets.

Physics first, ML second

A tight kinematic prior – the trapezoid template – beat black-box alternatives at every stage.

σ is a feature, not an output

Letting the model report its own uncertainty unlocked the quality filter, the CI, and downstream calibration.

Cross-phone is the real test

Numbers from a single phone are misleading. Four-phone simultaneous recording was decisive in catching biases.

Q & A

Thank you.

Eyal Yakir · Uriya Cohen-Eliya

Appendix

Detector Hyperparameter Walkthrough

Coarse grid then refine. Defaults chosen on the train fold; never re-tuned on test.

Parameter	Search range	Default	Source
τ_r ramp duration	0.5–1.5 s	0.9 s	<i>Ride dynamics</i>
τ_p plateau duration	0.0–6.0 s	1.5 s	<i>Lobe length</i>
a_0 plateau amplitude	0.4–1.4 m/s ²	0.9 m/s²	<i>Cabin spec</i>
τ_{thr} detection thresh.	2–6 σ	3σ	<i>Adaptive</i>
Δt_{pair} gap window	0.5–30 s	auto	<i>Per-ride bound</i>

Per-Experiment Detection Breakdown

Detection metrics by building. Beit Yitzchaki is the held-out test set.

Building	F1*	IoU-F1 @ 0.5
Millenium – Internal	0.84	0.69
Millenium – Outside	0.79	0.61
Acro	0.82	0.65
Beit Mansour	0.83	0.68
Barllan-2 (Herzliya)	0.78	0.59
Haari (Ramat Gan)	0.81	0.64
Beit Yitzchaki (HOLD-OUT)	0.875	0.830

Trapezoid σ : CRB Derivation

Cramer–Rao lower bound on $\sigma_{\Delta h}$ given the trapezoid Jacobian and white sensor noise.

$$y(t) = T(t; \theta) + \varepsilon(t), \quad \varepsilon \sim \mathcal{N}(0, \sigma_a^2 I)$$

$$I(\theta) = \frac{1}{\sigma_a^2} J(\theta)^\top J(\theta) \quad (\text{Fisher information})$$

$$\text{Cov}(\hat{\theta}) \geq I^{-1}(\theta) \quad (\text{Cramer–Rao})$$

$$\sigma_{\Delta h}^2 = \nabla_\theta \Delta h^\top I^{-1}(\theta) \nabla_\theta \Delta h \quad (\text{delta method})$$

$$\text{with } \nabla_\theta \Delta h = (\tau_p(\tau_r + \tau_p), a_0 \tau_p, a_0(\tau_r + 2\tau_p)).$$

ZUPT: Noise Budget

Three error sources dominate ZUPT estimates. Each scales differently with ride duration T .

Sensor noise

$$\sigma_h \propto \sigma_a T^{1.5} / \sqrt{N}$$

Reduced by sampling rate $\uparrow$ and stationary calibration.

Gravity bias

$$\sigma_h \propto \Delta g T^2$$

Worst at long rides.
Mitigated by per-ride gravity estimate.

Tilt error

$$\sigma_h \propto \sin(\delta\theta) g T^2$$

Mitigated by long stationary windows;
pocket motion remains a risk.

Ground-Truth Audit: Four-Phone Cross-Check

Each ride is recorded by four phones simultaneously. Cross-phone disagreement is our GT noise floor.

0.08 m

median cross-phone σ

99.2%

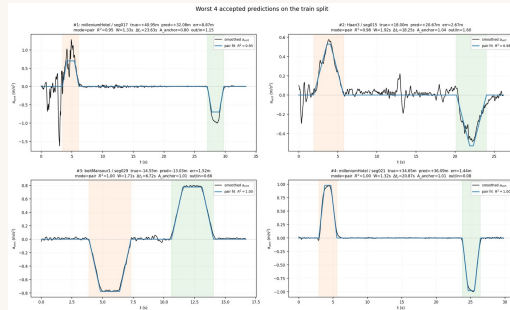
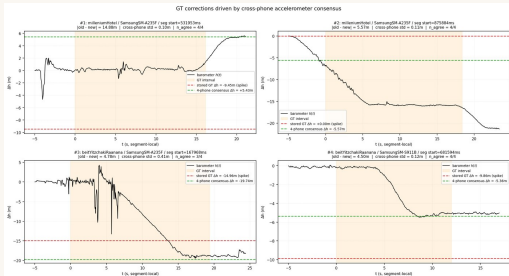
*rides agree within
 ± 0.5 m*

12 / 397

manual edits required

Where the barometer agrees with the trapezoid + ZUPT cross-check across all four phones, we trust the GT to ± 10 cm. The handful of edits address PRS dropouts.

Worst-Prediction Case Studies



Two failure cases. Left: short ride with overlapping lobes. Right: pocket reorient mid-ride.

Architecture / File Layout

Two-stage pipeline; each stage is a Pydantic-configured dispatcher.

<code>src/segmentation/</code>	Stage 1 – Segmenter dispatcher; matched-filter & pressure-filter algorithms.
<code>src/prediction/</code>	Stage 2 – Predictor dispatcher; barometer, ZUPT, trapezoid algorithms.
<code>src/utils/</code>	Stage-agnostic helpers – gravity removal, conformal calibrator, pulse fits.
<code>src/data/</code>	Loaders, GT editor, dataset definitions.
<code>src/pipelines/</code>	End-to-end Streamlit Boutique Pipeline.
<code>docs/latex/</code>	This presentation & report; LaTeX source + figures.